# Operations with Decimals

**Exam code:** 0607

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## Operations with decimals

### How do I add or subtract decimals without a calculator?

- If the numbers involve **decimals**, the column methods for addition and subtraction can be used
  - Line the place value columns up carefully
  - Make sure the **decimal points** are **all in the same column**
    - Writing zeros (often called place value holders) can help keep everything in line
  - e.g. 2.145 + 13.02 would be written as

$$\begin{array}{r} 02.145 \\ +\ 13.020 \\ \hline . \end{array}$$

### Worked Example

Johnny has £32.50 and spends £1.74.

Calculate how much money Johnny has left.

**Answer:**

This is a subtraction question as Johnny has *spent* money

Make a quick estimate

$$33 - 2 = 31$$

Align the digits by place value, ensuring £32.50 is the top number and using the decimal points as a starting point
Using place value holding zeros is optional

$$\begin{array}{r} 32.50 \\ -\ 01.74 \\ \hline . \end{array}$$

The column furthest right is the hundredths column
0 is smaller than 4 so borrow from the next (tenths) column

**Stuck on something?**

Ask questions and get help with exactly what you're revising.

Got it

$$\begin{array}{r} .\ \ 6 \end{array}$$

Next is the tenths column, but again 4 is smaller than 7 so borrow 10 from the ones column

$$\begin{array}{r} 1\ {}^{1}4 \\ 3\cancel{2}.\cancel{5}^{1}0 \\ -01.74 \\ \hline .\ \ 6 \end{array}$$

14 - 7 = 7 in the tenths column and continue working 'right to left'

$$\begin{array}{r} 1\ {}^{1}4 \\ 3\cancel{2}.\cancel{5}^{1}0 \\ -01.74 \\ \hline 30.76 \end{array}$$

Check the final answer is similar to the estimate
£31 and £30.76 are reasonably close

**Johnny has £30.76 left**

Help me understand

## How do I multiply decimals without a calculator?

- The standard methods for multiplication can be easily adapted for use with decimal numbers
  - E.g. column method, lattice method and grid method
- You can **make a problem easier** by converting to **integer values** then undoing the action to the answer
  - **Multiply** each value by a **power of 10** to make it an integer
  - Perform the easier multiplication
  - Undo the initial action by **dividing** by the **same powers of 10**
  - E.g. 2.5 x 4.01
    - Multiply 2.5 by 10 to give 25 and 4.01 by 100 to give 401
    - 25 x 401 = 10 025
    - Divide the answer by 10 and 100 to undo the initial changes
    - So the answer is 10.025
- An alternative method is to **ignore** the **decimal point** whilst multiplying
  - then **put it back in** the correct place for the final answer using **estimation**
  - E.g. 1.3 x 2.3
    - Ignoring the decimals this is 13 x 23, which works out to 299
    - 1.3 x 2.3 is approximately 1.5 x 2, which is 3
    - So we know the answer should be close to 3, rather than 0.3 or 30
    - So the answer is 2.99

A good way to check your answer without a calculator is to estimate it by rounding everything to 1 or 2 significant figures.

## Worked Example

Without using a calculator, find $1.57 \times 0.78$ .

**Answer:**

*Method 1*

Multiply both values by powers of 10 to create a calculation using integer values

$$1.57 \times 100 = 157$$
$$0.78 \times 100 = 78$$

$$157 \times 78$$

Using the grid method

| | 100 | 50 | 7 |
|---|---|---|---|
| 70 | 7000 | 3500 | 490 |
| 8 | 800 | 400 | 56 |

$$\begin{array}{r} 7000 \\ +3500 \\ 490 \\ 800 \\ 400 \\ 56 \\ \hline 12246 \end{array}$$

$$157 \times 78 = 12\,246$$

Undo the initial action by dividing the answer by 10 000 (This is dividing by 100 and then 100 again)

$$12\,246 \div 10\,000$$

**1.2246**

*Method 2*

Ignore decimal point to form a different, non-decimal calculation

$$157 \times 78$$

Using the grid method

| | 100 | 50 | 7 |
|---|---|---|---|
| 70 | 7000 | 3500 | 490 |
| 8 | 800 | 400 | 56 |

$$\begin{array}{r} 800 \\ 400 \\ 56 \\ \hline 12246 \end{array}$$

157 × 78 = 12 246

Estimate the original calculation to write this as the correct answer

1.57 × 0.78 is approximately 2 × 1 = 2

So the correct answer is between 1 and 10

**1.2246**

## How do I divide a decimal without a calculator?

- Use a similar approach to when multiplying decimals
- **Make a problem easier** by converting to **integer values** then undoing the action to the answer
  - Write the division as a **fraction**
  - **Multiply** each value by a **power of 10** to make it an integer
  - Perform the easier division
  - Undo the initial action
    - **divide** by the **same power of 10** the **numerator** was multiplied by
    - **multiply** by the **same power of 10** the **denominator** was multiplied by
  - E.g. 37.5 ÷ 0.25
    - $37.5 \div 0.25 \rightarrow \frac{37.5}{0.25}$
    - $\frac{37.5 \times 10}{0.25 \times 100} = \frac{375}{25}$
    - $\frac{375}{25} = 15$
    - Undo the initial actions $\frac{15 \times 100}{10}$
    - So the answer is 150
- Alternatively, ignore the decimal point whilst carrying out the division
  - then **put it back in** the correct place for the final answer using **estimation**
  - E.g. 4.68 ÷ 6
    - 4.68 ÷ 6 is approximately 5 ÷ 6, which is between 1 and 0.5
    - So we know the answer should be 0.78, rather than 7.8 or 0.078

### Worked Example

Without using a calculator, find 69.02 ÷ 0.7 .

Write as a fraction

$$\frac{69.02}{0.7}$$

Multiply both numbers by powers of 10 to create a calculation with integer values

$$\frac{69.02 \times 100}{0.7 \times 10} = \frac{6902}{7}$$

Use short division (bus stop method)

$$\begin{array}{r} 0\ 9\ 8\ 6 \\ 7\overline{)6^{6}9^{6}0^{4}2} \end{array}$$

$$\frac{6902}{7} = 986$$

Convert the answer
Divide by 100 to cancel out 69.02 being multiplied by 100
Multiply by 10 to cancel out 0.7 being multiplied by 10

$$\frac{986 \times 10}{100}$$

**98.6**

*Method 2*

Ignore the decimal point to form a different, non-decimal calculation

$$6902 \div 7$$

Use short division (bus stop method)

$$\begin{array}{r} 0\ 9\ 8\ 6 \\ 7\overline{)6^{6}9^{6}0^{4}2} \end{array}$$

$$6902 \div 7 = 986$$

Estimate the original calculation to write this as the correct answer

69.02 ÷ 0.7 is approximately 70 ÷ 1 = 70

So the correct answer is between 10 and 100

**98.6**

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